Osiris

Design System & Mockup Reference

Marine analytics platform · v1

All eight locked screens, design foundations, and patterns extracted.

Prepared for Vineet · May 2026

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1. Foundations

Visual direction, typography, color and spacing

Visual direction

Modern-SaaS leaning operational. Decisions and primary controls receive SaaS chrome — generous spacing, clear hierarchy, recommendation-forward. Data surfaces stay denser, with tabular figures and tight rows for scanning. The product is designed for a recommendation-with-human-confirmation model now, with a clear path to greater autonomy as confidence in the engine grows.

Reference points: Linear's chrome and typography rhythm; Vercel's operational-state handling; Cursor and GitHub Copilot for the human-in-the-loop confirmation pattern.

Color

Token	Hex	Use
Background — primary	`#ffffff`	Page surfaces, card interiors
Background — secondary	`#f5f5f7`	Status strips, muted rows, past segments
Background — tertiary	`#e8e8ec`	Map land masses, faint dividers
Background — info	`#eaf2fc`	Recommended profile, hero KPI, current segment
Text — primary	`#18181b`	Body content, emphasis
Text — secondary	`#5c5c66`	Supporting text, metadata
Text — tertiary	`#9999a3`	Labels, dates, captions
Accent — info (blue)	`#2563eb`	Brand color, primary actions, ‘Now’
Accent — success (green)	`#059669`	Gains, light congestion
Accent — warning (amber)	`#d97706`	Anchored, moderate congestion

Accent — danger (red)	`#dc2626`	Losses, heavy congestion
Border — tertiary	`#e1e1e6`	Default card and row borders

Typography

Inter (with system-UI fallback) for all UI text. Tabular figures (font-variant-numeric: tabular-nums) wherever numbers appear vertically aligned in lists, columns, or stats. Sizes range 10px (labels) through 32px (hero KPI values). Weight typically 400 with 500 used for emphasis; bold (700) is rare. Letter-spacing tightened slightly (-0.01em) on large display titles.

Spacing & layout

12px gutters between cards in dense regions, 18px between page sections. Card padding 14–18px depending on density. Border-radius: 6px for inputs and chips, 10px for cards and rows. 0.5px hairline borders throughout — thicker borders (1.5px) reserved for the recommended profile and conclusion accents. Generous around primary actions; tighter inside data surfaces.

2. Honesty patterns

How Osiris earns trust on every screen

Each screen carries at least one mechanism that helps a sharp viewer trust the numbers. These patterns compose: the same vocabulary appears across screens with consistent meaning, so users learn them once.

Coverage badges

Per-row pill on Fleet indicating prediction availability. Three states: 'Predictions available' (green), 'Limited coverage' (amber), 'Type pending' (gray). Renders gracefully when limits apply, looks identical when they don’t. Supports the ‘data event, not UI event’ property — backend coverage expansion changes the badge, not the UI.

Source labels

Every speed-schedule segment carries a label telling its provenance: ‘AIS observed’ (fact), ‘Predicted’ (model output), or ‘Recommended’ (system suggestion). This is what keeps the UI honest as the product moves toward greater autonomy.

Inline confidence ranges

Predictions show their uncertainty inline: ‘Tomorrow 14:30 ± 2h’, ‘5h 30m ± 1h 30m’. Pulled from the model’s per-bucket error distribution at request time, so as the model improves the bands tighten on their own.

Methodology caveat

On Historical Analytics vessel summary, a small but visible paragraph between KPIs and distribution: ‘Recommendations regenerated by the production engine using only data available at each voyage’s decision point. No hindsight. Fuel and cost are modelled estimates; assumptions: $620/t bunker, $2,400/h port time.’

Wins-and-losses distribution

Voyage outcomes always shown as three buckets: improved / no material change / worsened. A demo that hides losses reads as cherry-picking; this version exposes them on the summary card and includes a real loss row in the voyage list.

Calibration receipts

Voyage detail leads with a 3-card grid showing predicted vs. actual on arrival, anchorage, and berth, with the error called out. The engine’s claims meet ground truth before any savings claim is made.

Observed-vs-predicted timeline

Port Intelligence shows two parallel rows of congestion bands: observed (solid, past only) and predicted (faded, full range). Where they overlap, the user can compare model calls to reality. Earns trust on past, spends it on future.

Per-vessel actual + saved side by side

Voyage rows on Historical Analytics show the saved figure as the primary value and the actual figure beneath it as context. No headline saving floats alone; every claim is paired with what was actually spent.

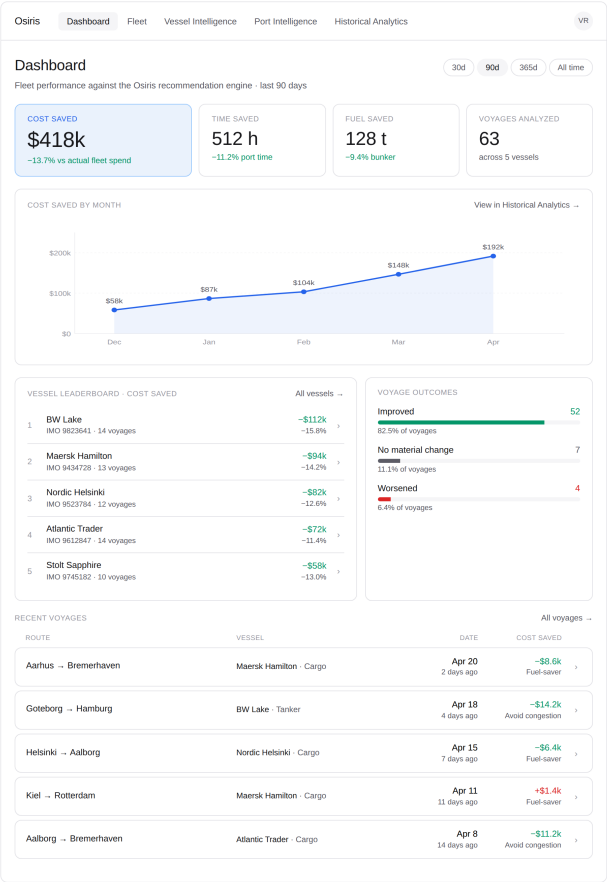
3. Screens

In user-flow order — Dashboard, Fleet, Vessel Intelligence, Port Intelligence (selector and detail), Historical Analytics (three levels)

Each screen below is the final locked iteration. Earlier versions and the design conversations that produced them are not included.

3.1 Dashboard

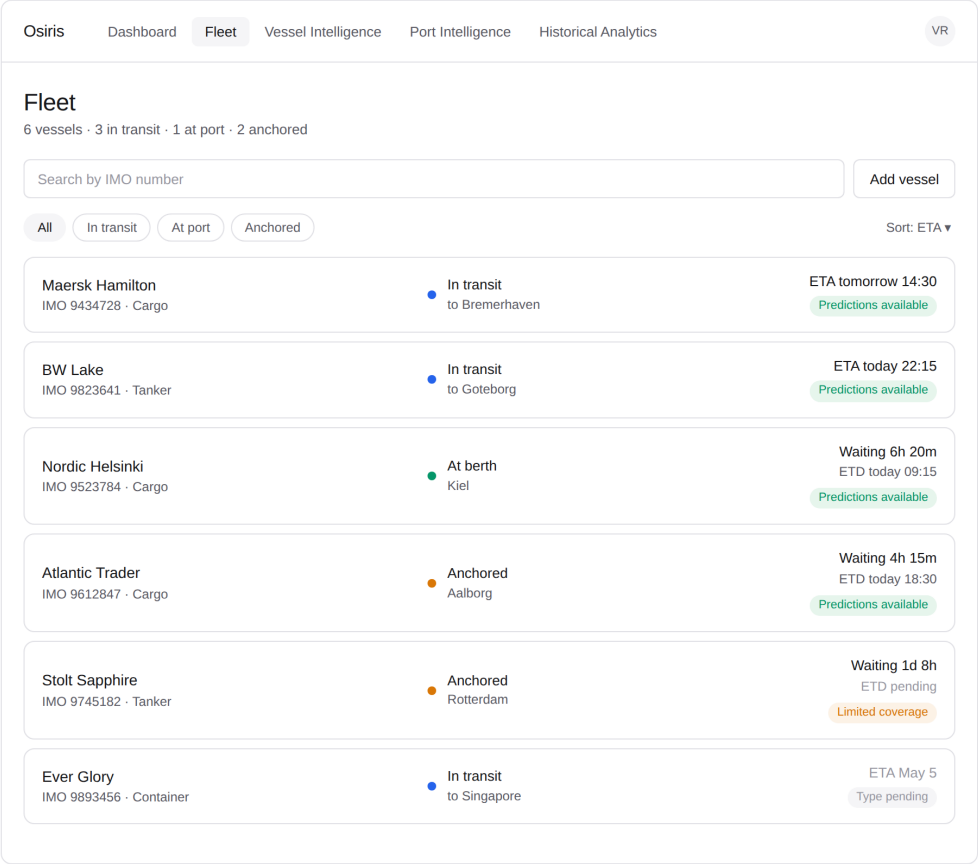
Executive view — fleet performance against the Osiris recommendation engine



Time scope is the only top-level control (30d / 90d / 365d / all). **Cost** is the hero KPI; time and fuel sit as supporting receipts. The savings trend chart, vessel leaderboard, and recent-voyages strip all link to Historical Analytics so any aggregate number can be defended.

3.2 Fleet

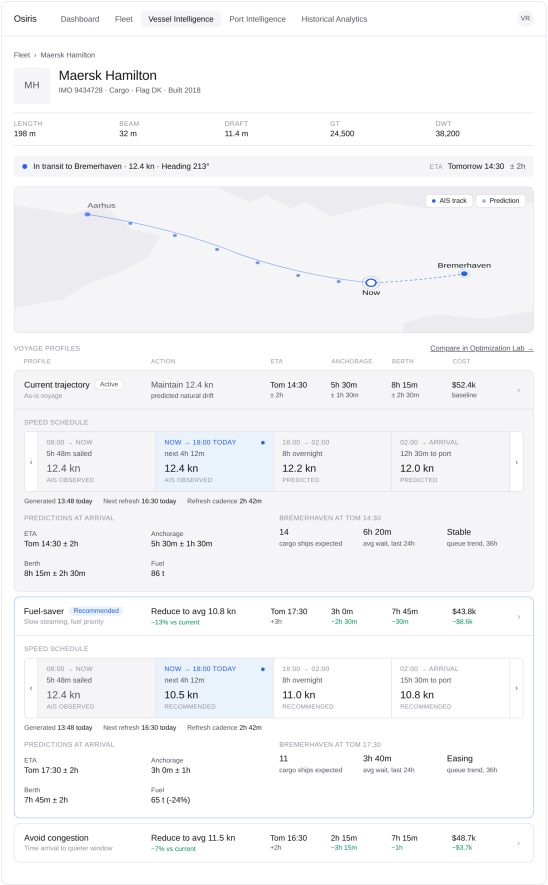
Operational view of the curated fleet — status, destination, prediction coverage at a glance



Variable row heights signal vessel state — in-transit rows are compact, at-port and anchored rows are taller with waiting time and ETD. The coverage badge on each row makes prediction availability visible without explanation.

3.3 Vessel Intelligence

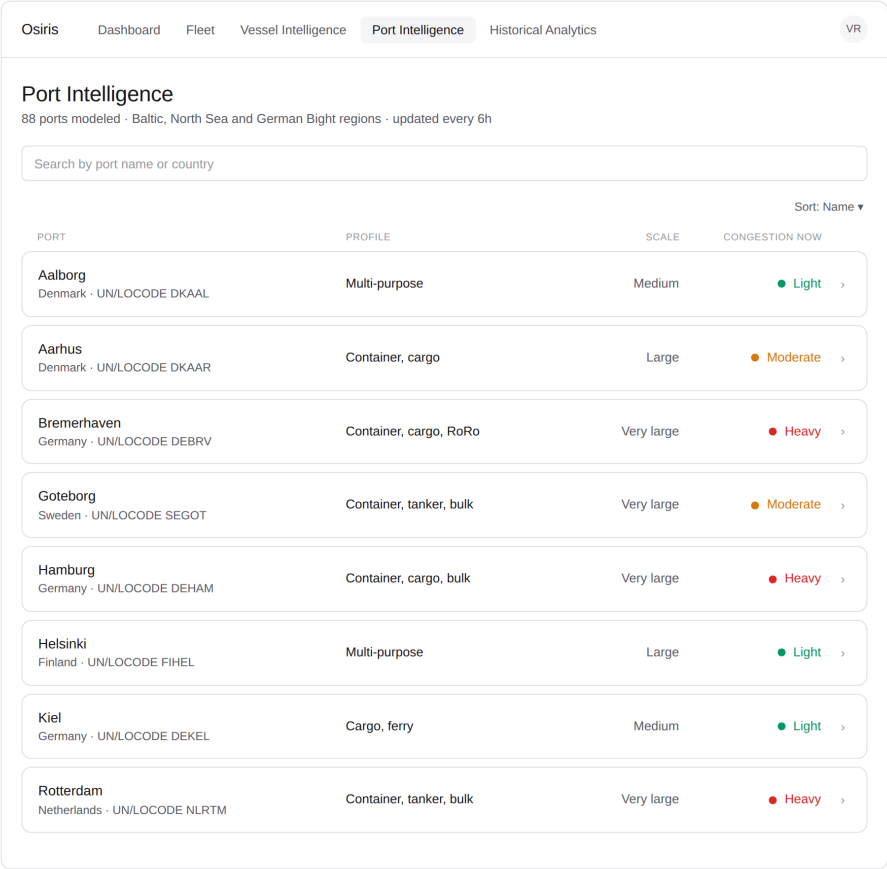
Single-vessel decision support — current voyage, profile comparison, time-segmented speed schedule



The speed schedule strip is the page's key innovation: 4 segments (1 past, 1 current, 2 future) with scroll arrows for longer voyages, and source labels on each. Both Current and Recommended profiles expand to show predicted port outcomes plus the destination's congestion forecast at the predicted arrival.

3.4 Port Intelligence — selector

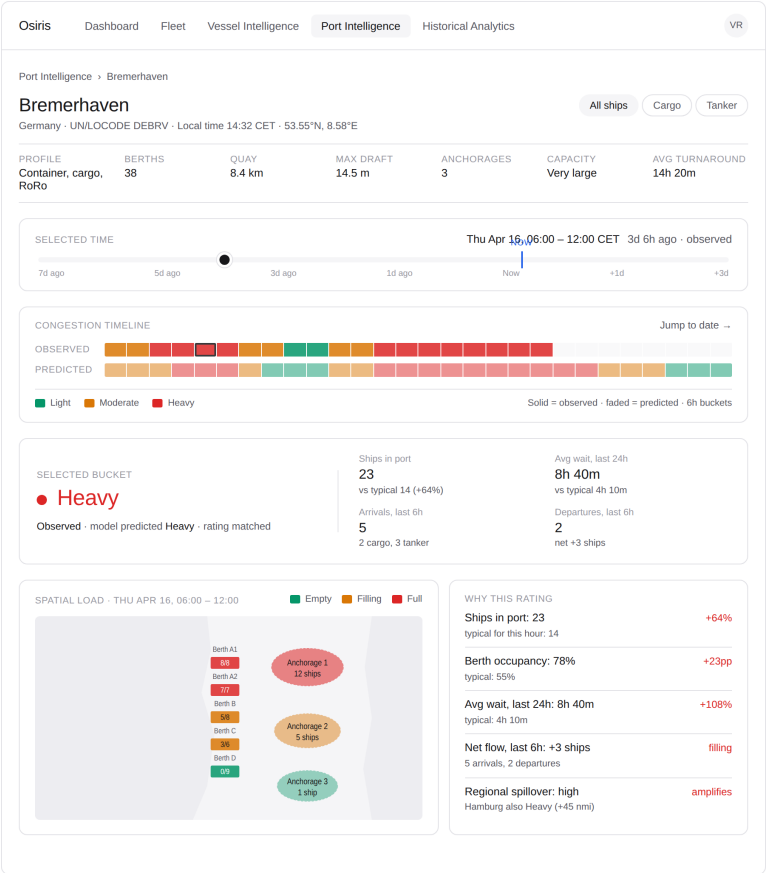
Searchable list of all 88 modeled ports with current congestion at a glance



Same chrome as the Fleet selector, different semantic load. Profile column shows the multi-cargo nature of real ports. Current congestion is the only live dimension on an otherwise static page.

3.5 Port Intelligence — detail

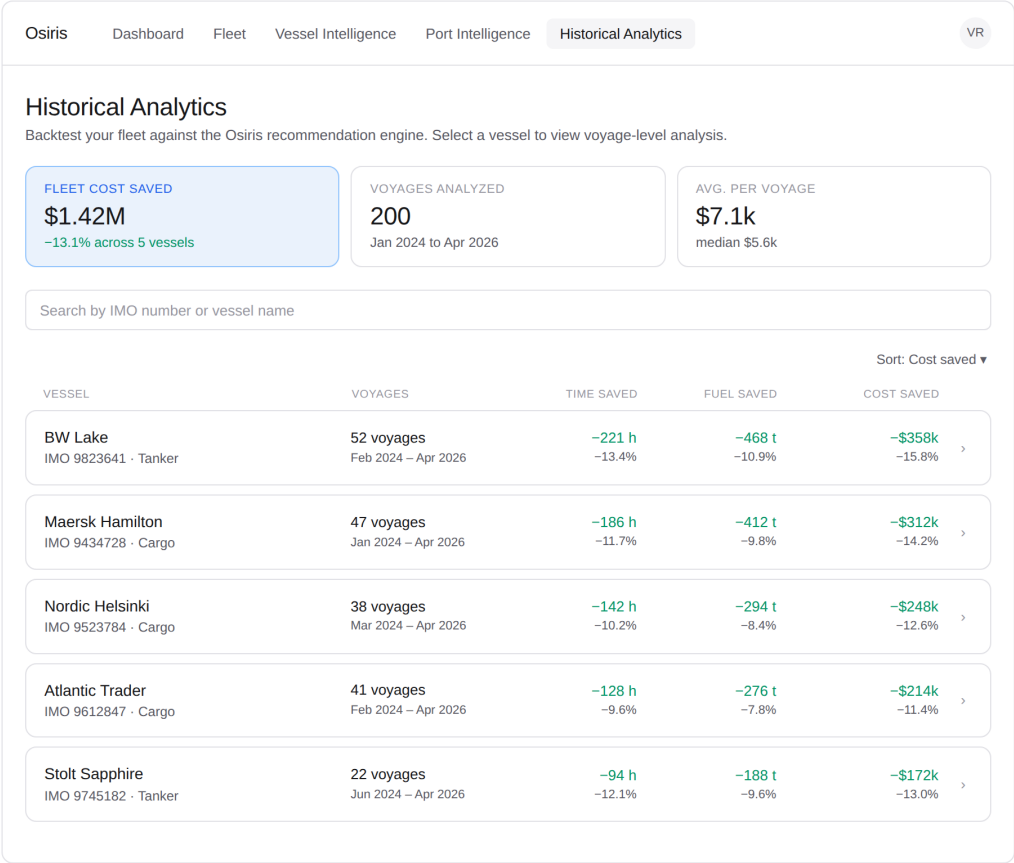
Time-aware port view with observed-vs-predicted timeline, headline rating, spatial heatmap, and feature attribution



The two-row timeline strip is the trust loop in concrete form — observed (solid) and predicted (faded) bars sit parallel so the user can scan for disagreement. The schematic spatial heatmap colors berth and anchorage zones by load. The 'why this rating' panel surfaces feature contributions ranked by anomaly.

3.6 Historical Analytics — landing

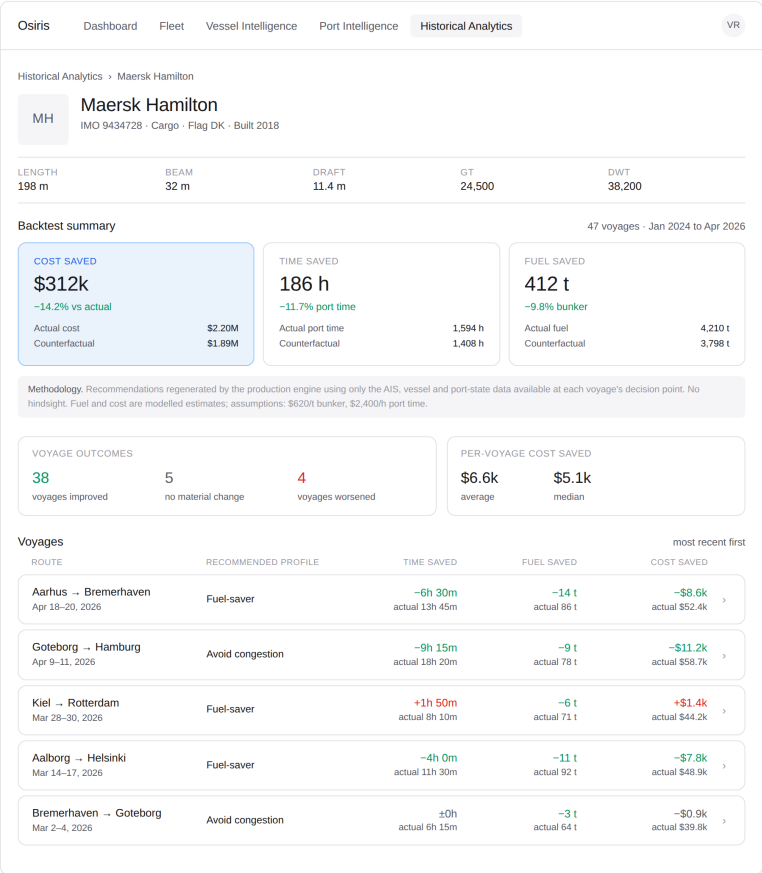
Fleet roll-up KPIs and per-vessel savings rows — the demo opener



Three aggregate KPIs at the top frame the value claim. Per-vessel rows show all three savings dimensions (time / fuel / cost) with percentage subtitles for cross-vessel comparison.

3.7 Historical Analytics — vessel summary

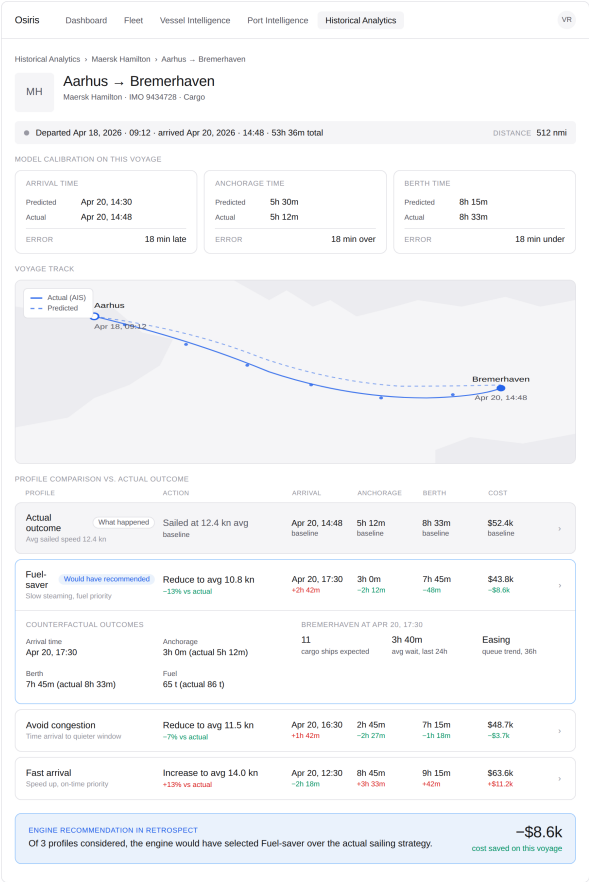
Backtest summary for one vessel — aggregate savings, voyage outcome distribution, voyage list



Hero/secondary KPI treatment: cost saved is the headline, time and fuel are supporting evidence. Methodology caveat sits between KPIs and distribution as honest disclosure. Voyage list includes both wins and losses.

3.8 Historical Analytics — voyage detail

Single voyage backtest — calibration receipts, actual-vs-predicted track, profile comparison, retrospective recommendation



Model calibration card sits before the map and answers ‘did our predictions land?’ before showing any savings claim. The actual outcome is itself a profile row at the top of the comparison; counterfactuals are deltas against it. Recommendation conclusion is a tinted footer card — quiet but unambiguous.

4. Component vocabulary

Reusable patterns that appear across multiple screens

Top navigation

Five tabs: Dashboard, Fleet, Vessel Intelligence, Port Intelligence, Historical Analytics. Active tab uses the secondary background tint with primary text color and 500 weight. Identical across all pages.

Breadcrumb

Small secondary-text strip above the page title, with separator chevrons. Drill-down levels: ‘Fleet › Maersk Hamilton’, ‘Historical Analytics › Maersk Hamilton › Aarhus → Bremerhaven’. Every breadcrumb level is clickable.

Vessel header

Monogram thumbnail (60px), name as h1, IMO + ship type + flag + year metadata line. Used identically on Vessel Intelligence and Historical vessel summary.

Specs strip

Horizontal grid of label-value pairs with hairline top and bottom borders. Five fields for vessels (Length, Beam, Draft, GT, DWT), seven for ports (Profile, Berths, Quay, Max draft, Anchorages, Capacity, Avg turnaround). Labels in uppercase tertiary text, values in primary text with tabular figures.

Status strip

Tinted single-line strip (secondary background) with colored dot, status text, and key live data. Used for ‘In transit to Bremerhaven’ on Vessel Intelligence and as voyage facts on Voyage Detail.

Profile row

The fundamental unit of decision presentation. Columns: profile name + tag, action with %% delta, ETA, anchorage, berth, cost. Active baseline uses tinted background; recommended uses 1.5px info border. Expandable to show speed schedule and predictions.

Speed schedule strip

4-segment horizontally-scrollable strip with left/right arrows. Each segment shows time range, duration, speed, and source label. Past = secondary tint, Now = info tint with pulse dot, future = default. Used for both Current and Recommended profiles.

Coverage badge

Inline pill: ‘Predictions available’ (success), ‘Limited coverage’ (warning), ‘Type pending’ (neutral). Renders the state of model support per-row without changing layout.

KPI card

Padded card with uppercase tertiary label, large primary value (24–32px tabular), and a percentage subtitle. Hero variant uses info background. Secondary KPIs may include ‘actual vs counterfactual’ data lines.

Distribution panel

Three rows showing improved / no material change / worsened counts with horizontal bars sized by percentage. Used on Voyage outcomes (vessel summary and dashboard).

Congestion bands

Two-row horizontal strip of colored boxes representing 6h time buckets. Top row observed (full opacity), bottom row predicted (50%% opacity). Three states: light/moderate/heavy + empty for no-data.

Spatial heatmap

Schematic port map with berth zones (rectangles) and anchorage zones (ellipses) colored by load. Berth zones show occupancy fraction (e.g. ‘7/8’); anchorages show ship count.

Feature attribution panel

Vertical list of feature contributions to a model rating. Each row has the feature name + value, the typical baseline, and a directional indicator on the right.

Time slider

6-pixel track with current-time anchor (‘Now’ vertical line in info color), draggable handle, and tick scale below (‘7d ago’ through ‘+3d’). 6h bucket snap behavior.

Methodology caveat

Quiet single-paragraph disclosure with secondary-tint background, rendered between KPIs and supporting data. Discloses backtest type, fuel/cost assumptions, and any other model limitations relevant to the page’s claims.

5. Deferred for v2

What was deliberately left out of v1, in priority order

Recommendation engine

The largest gap. Profile recommendations on Vessel Intelligence and the retrospective recommendation on Voyage Detail are placeholders. v1 design assumes an argmin-over-profiles selection; the actual engine evaluating fuel + waiting penalty + congestion needs to be built. Until it exists, the ‘Recommended’ badge is unsupported.

Fuel model

All fuel and cost estimates currently assume a cubic-speed-law approximation labeled as a modeling assumption. Per-vessel fuel curves (from class society data, ship specs, or AIS-derived) would tighten the numbers. The UI structure for this exists; only the underlying calculation needs improvement.

Co-Pilot panel

The human-in-the-loop confirmation surface that turns a recommendation into a confirmable plan. Designed conceptually but deliberately deferred: building it as a stub would weaken the demo. Comes back when the recommendation engine is real, with the existing design language as the foundation.

Optimization Lab

User-driven profile construction surface where operators define new strategies (e.g. weight fuel 60%, wait 30%, on-time 10%). v1 ships with canned profiles only; the Lab is where the operator becomes the system designer. Inherits the speed-schedule and profile-card patterns from Vessel Intelligence.

Alerts and the agentic loop

Background monitoring that re-evaluates plans every 2–6h and surfaces ‘recommendation has changed’ events. Most of the autonomy-arc work lives here. v1 is read-only; v2 begins the push to autonomy with a notification surface.

Real Google Maps integration

Current voyage-track and port spatial-heatmap visualizations use schematic SVG. Production wiring through Google Maps JavaScript API would render geographic accuracy and per-point timestamp tooltips on hover.

Coverage expansion

v1 covers 88 ports across the Baltic, North Sea, and German Bight, plus Cargo and Tanker ship types. These are data limitations not product limitations — the design renders gracefully as coverage expands. Backend API for /coverage drives this dynamically; no UI work required as new ports and types come online.

Historical Analytics — predicted-vs-actual logging

The voyage-level calibration card shown in v1 is real. A fleet-level prediction-error history chart would extend this to a longitudinal view (‘model accuracy by month’, ‘feature importance over time’). Useful for both customer trust and internal model tuning.

End of document